

Ryan Jan

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EDUCATION

California Polytechnic State University

Bachelor of Science in Electrical Engineering, Minor in Mathematics | GPA: 3.67/4.00

San Luis Obispo, CA

Expected June 2029

ROBOTICS EXPERIENCE

FIRST Tech Challenge Robotics – Ramifications #5155

St. Louis, MO

Programming Lead (2022–2025) & CAD/Design Lead (2023–2025) | [GitHub](#)

Aug. 2021 – May 2025

- Led programming and computer-aided design (CAD) for a 10-member team, directly mentoring 3 students in Java, control systems, and Onshape.
- Authored 3,000+ lines of custom Java across 50+ source files, including a 25-class command framework and four autonomous routines using the FTC software development kit (SDK).
- Implemented field-oriented mecanum drive and closed-loop X/Y position and heading control with goBILDA Pinpoint localization; used an existing JavaFX simulator for off-robot autonomous testing.
- Directed Onshape design and fabrication through computer numerical control (CNC) machining, 3D printing, and laser cutting; the team's control software received FIRST Control Awards in two consecutive seasons.

SELECTED ENGINEERING PROJECTS

Two-Wheel Self-Balancing Robot | *Java, FTC SDK, Onshape* | [GitHub](#)

- Individually designed and fabricated a balancing prototype using a REV Control Hub, BHI260AP inertial measurement unit (IMU), motor encoders, and a custom 3D-printed chassis.
- Implemented and compared proportional-integral-derivative (PID) control with a four-state linear-quadratic regulator (LQR) using pitch angle, pitch rate, wheel position, and wheel velocity.
- Modeled the robot as an inverted pendulum and solved the algebraic Riccati equation in Java to compute full-state feedback gains; evaluated both controllers with FTC Dashboard telemetry.

Small C Compiler | *C, x86-64 Assembly, Git* | [GitHub](#)

- Built a 2,800+ line C compiler with lexical analysis, recursive-descent parsing, abstract syntax tree construction, type and scope analysis, and direct x86-64 assembly generation.
- Implemented a substantial C subset covering functions and recursion, control flow, pointers and arrays, strings, aggregate types, enumerations, type definitions, and casts.
- Validated generated programs with 362 assertions across 15 automated test programs using Make and the GNU Compiler Collection (GCC) to preprocess, assemble, link, and execute each case.

IR-Controlled Temperature Regulator | *ATmega2560, Arduino, Tinkercad* | [GitHub](#)

- Built an individual fan-control system using a negative temperature coefficient (NTC) thermistor voltage divider, PN2222 transistor driver, flyback diode, status LED, and infrared remote.
- Converted 10-bit analog-to-digital converter (ADC) readings to temperature with a Steinhart–Hart model and actuated the fan above a 71-degree Fahrenheit setpoint; verified operation through Tinkercad simulation and physical hardware testing.

TECHNICAL SKILLS

Languages: Java, C, C++, Python, MATLAB, Bash, x86-64 Assembly

Embedded & Controls: FTC SDK, Arduino, PID and LQR control, motor control, sensor integration

Development & Design: Git, Linux, Android Studio, JavaFX, Onshape, LTspice

Lab & Fabrication: Oscilloscope, multimeter, function generator, power supply, soldering, CNC machining, 3D printing, laser cutting